

Lab 10: Epigenetics — Gene Regulation

BIOL-1

Name: _____ Date: _____

Overview

Your cells share the same genome, but a brain cell and a muscle cell use different subsets of genes.

Gene regulation is how cells turn genes on or off. **Epigenetics** describes changes in gene expression **without changing the DNA sequence**—for example, chemical tags on DNA or on the proteins that package it.

This lab covers bacterial gene control (**lac operon**), human **epigenetic** mechanisms (DNA methylation, histone modification), **X-inactivation**, and ways the **environment** can affect which genes stay active.

Learning Objectives

By the end of this lab, you will be able to:

1. Explain why gene regulation is essential for multicellular organisms.
2. Describe how the lac operon turns genes on and off in bacteria.
3. Explain how DNA methylation and histone acetylation control gene expression.
4. Describe X-inactivation and its visible effects (calico cats).
5. Give examples of how environment influences gene expression.

Materials

- This worksheet
 - Colored pencils (optional)
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Part 1: Why Gene Regulation Matters

A liver cell and a neuron share the same DNA. They differ because they express different genes and therefore make different proteins.

1. Do all cells in your body contain the same DNA?

2. Do all cells in your body produce the same proteins?

3. If a muscle cell and a nerve cell have the same DNA, how can they look and function so differently?

Part 2: The Lac Operon — Gene Regulation in Bacteria

E. coli can digest lactose, but building enzymes costs energy. The lac operon is an ON/OFF switch: make those enzymes only when lactose is available.

How the lac operon works:

- **Lactose ABSENT:** Repressor binds the operator; RNA polymerase is blocked; enzyme genes stay **OFF**.
- **Lactose PRESENT:** Lactose binds the repressor; repressor releases the operator; RNA polymerase transcribes the genes; enzymes are made.

4. When lactose is absent, are the lac enzyme genes ON or OFF?

5. What is the job of the repressor protein in the lac operon?

6. What happens to the repressor protein when lactose enters the cell?

7. Why is the lac operon energy-efficient for the bacterium?

8. Prediction: If a mutation destroys the repressor gene (so no repressor is made), will the lac operon genes be always ON, always OFF, or normal?

9. Prediction: If a mutation in the promoter prevents RNA polymerase from binding, will the genes be always ON, always OFF, or normal?

Part 3: Epigenetic Mechanisms

Epigenetic marks sit on DNA and histones. They make genes easier or harder to read without changing the DNA sequence.

10. Does epigenetics change the order of DNA bases (the DNA sequence)? Answer yes or no.

11. The proteins that DNA wraps around are called _____.

12. Which packaging is easier for RNA polymerase to read—loose or tight?

13. DNA methylation — In one sentence, what is it?

Typically turns genes ON or OFF?

14. Histone acetylation — In one sentence, what is it?

Typically turns genes ON or OFF?

15. In the table below, circle one choice in each cell (one for DNA packaging, one for gene expression in each row).

Chromatin Type	DNA Packaging	Gene Expression
Euchromatin	Loose / Tight	ON / OFF
Heterochromatin	Loose / Tight	ON / OFF

16. In a skin cell, the gene for keratin (a skin protein) is actively making protein. Is the keratin gene packaged as euchromatin or heterochromatin in this cell?

17. In that SAME skin cell, the gene for hemoglobin (a blood protein) is completely silent. Is the hemoglobin gene packaged as euchromatin or heterochromatin?

18. Adding acetyl groups to histones associated with a gene tends to turn that gene's expression up or down?

Part 4: X-Inactivation — Epigenetics You Can See

Females (XX) inactivate one X in each cell early in development, at random. Some cell patches use the maternal X; others use the paternal X.

19. What is X-inactivation?

20. What is the condensed, inactivated X chromosome called?

21. Calico cats have patches of orange and black fur. The gene for coat color is on the X chromosome. Using what you know about X-inactivation, explain why calico cats have a patchy pattern instead of a uniform color.

22. Why are calico cats almost always female?

Part 5: Environment and Epigenetics

You cannot change your DNA sequence, but lifestyle and environment can shift which genes are active through epigenetic marks.

23. Give two examples of environmental influences on epigenetics (for each, add one short note on how it might matter).

1.

2.

24. Identical twins share the same DNA sequence. As they age, they may differ in health, body type, and traits. *Using epigenetics and gene regulation*, explain how that is possible.

25. A tumor suppressor gene is heavily methylated but has *no* mutations in its DNA sequence. Could that contribute to cancer? Explain briefly.